

Hoonki Yeo

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RESEARCH INTERESTS

Optimization algorithms under uncertainty; optimization for machine learning; optimization for operational decision-making.

EDUCATION

Georgia Institute of Technology

Aug 2025 - Dec 2026 (expected)

M.S. in Operations Research, H. Milton Stewart School of Industrial and Systems Engineering

Atlanta, GA

- **Selected coursework:** Convex Optimization, Linear Optimization, Simulation, Theoretical Statistics, High-Dimensional Data Analytics, Computational Data Analytics.

University of Wisconsin-Madison

Aug 2019 - Aug 2022

B.S. in Data Science

Madison, WI

- **GPA:** 3.84/4.00. Graduated with Distinction; Dean's List for four semesters.

RESEARCH EXPERIENCE

Derivative-Free Stochastic Optimization for Constrained Learning

Mar 2026 - Present

Georgia Institute of Technology; Advisor: Prof. Sen Na

Atlanta, GA

- Formulated a partially linear single-index learning problem as stochastic optimization over a unit-norm index direction, with objective evaluations obtained through stochastic neural-network training.
- Implemented and verified the first-order variant of derivative-free stochastic sequential quadratic programming (DF-SSQP), including gradient estimation, moving averages, and constrained SQP updates.
- Designed and conducted controlled experiments comparing analytic and estimated constraint Jacobians and alternative perturbation distributions, assessing development loss, constraint violation, and variability across stochastic runs.

Multimodal Accessibility for a 20-Minute City in Tempe, Arizona

Sep 2026 - Present

Data4Good Analytics Competition · Team Research Project

- Designing a comparative study of walking, low-stress cycling, and public transit accessibility to daily destinations using twenty-minute coverage and travel-time measures.
- Reviewing network, transit-schedule, and destination data to design constrained routing analyses of how connectivity, cycling stress, and service conditions limit accessibility.

PROFESSIONAL EXPERIENCE

Data Scientist

2024 - 2025

Aivelabs

Seoul, South Korea

- Developed and deployed production machine learning workflows for customer value forecasting, from data preparation and model training to scoring.
- Deployed personalized recommendations and a multi-armed bandit feedback loop for outreach targeting.
- Developed customer segmentation and KPI dashboards to support marketing and sales decisions in collaboration with cross-functional teams.

Data Analyst Intern

2022 - 2023

Retentics

Seoul, South Korea

- Developed an internal Python preprocessing package to standardize data preparation workflows.
- Maintained data-processing and validation pipelines and conducted analyses for client projects to support operational decisions.

TEACHING EXPERIENCE

Peer Tutor

2020 - 2021

University of Wisconsin-Madison

Madison, WI

- Provided one-on-one tutoring in introductory statistics and probability for STAT 301 and STAT 309.

LEADERSHIP & SERVICE

Sergeant, Thermal Observation Device Operator

2016 - 2018

Republic of Korea Army

Yeoncheon, South Korea

- Led a small team responsible for observation-equipment readiness and training new personnel.

TECHNICAL SKILLS

Programming: Python, R, SQL.

Optimization: Gurobi, OR-Tools.

Scientific computing and modeling: NumPy, PyTorch, scikit-learn, TensorFlow.

Data systems: Apache Airflow, Apache Spark, MySQL, PostgreSQL, Tableau, AWS.

Research tools: Git, LaTeX.